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ORIGINAL ARTICLE

Mechanical and metabolic power in accelerated running–Part II: team sports

Cristian Osgnach[1] · Pietro E. di Prampero[1,2] · Paola Zamparo[3] · Jean-Benoit Morin[4] · Gaspare Pavei[5]

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Abstract

Purpose This manuscript is devoted to discuss the interplay between velocity and acceleration in setting metabolic and mechanical power in team sports.

Methods To this aim, an essential step is to assess the individual Acceleration-Speed Profile (ASP) by appropriately analysing training sessions or matches. This allows one to estimate maximal mechanical and metabolic power, including that for running at constant speed, and hence to determine individual thresholds thereof.

Results Several approaches are described and the results, as obtained from 38 official matches of one team (Italian Serie B, season 2020–2021), are reported and discussed. The number of events in which the external mechanical power exceeded 80% of that estimated from the subject's ASP (#*burst*) was 1.61 times larger than the number of accelerations above 2.5 m s^{-2} (#*acc*). The difference was largest for midfielders and smallest for attackers (2.30 and 1.36 times, respectively) due to (i) a higher starting velocity for midfielders and (ii) a higher external peak power for attackers in performing #*burst*. From the energetic perspective, the duration and the corresponding metabolic power of high-demanding phases (#*MPE*) were essentially constant (6 s and 22 W kg^{-1}, respectively) from the beginning to the end of the match, even if their number decreased from 28 in the first to 21 in the last 15-min period, as a consequence of the increased recovery time between #*MPE* from 26 s in the first to 37 s in the last 15-min period.

Conclusion These data underline the flaws of acceleration counting above fixed thresholds.

Keywords Acceleration · High intensity · Mechanical power · Metabolic power · Soccer · Performance analysis

Abbreviations

#*acc*	Number of intense acceleration actions
#*burst*	Number of intense mechanical actions
#*MPE*	Number of intense metabolic actions (Metabolic Power Events)
#*speed*	Number of intense speed actions
Δv_t	Increase of velocity at time t
τ	Time constant of the ASP
a	Acceleration
a_t	Instantaneous acceleration
a_0	Y-intercept of the individual ASP
a_{thr}	Acceleration threshold
AnE	Energy derived from anaerobic sources
ASP	Acceleration-Speed Profile
ATT	Attackers
CB	Centre back
CI	Confidence intervals
CM	Centre midfielder
$D_{acc>2.5}$	Distance covered when $a > 2.5$ m s^{-2}
$D_{MP>20}$	Distance covered when $MP > 20$ W kg^{-1}
$D_{speed>5.5}$	Distance covered when $v > 5.5$ m s^{-1}
$D_{speed>7.0}$	Distance covered when $v > 7.0$ m s^{-1}
DEF	Defenders
EE	Energy expenditure
EP	External power

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✉ Cristian Osgnach
cristian.osgnach@gmail.com

1 Department of Sport Science, Exelio srl, Udine, Italy

2 Emeritus Professor of Physiology, University of Udine, Udine, Italy

3 Department of Neurosciences, Biomedicine and Movement Sciences, University of Verona, Verona, Italy

4 Laboratoire Interuniversitaire de Biologie de la Motricité, Université Jean Monnet Saint-Etienne, Lyon 1, Université Savoie Mont-Blanc, Saint-Etienne, France

5 Department of Pathophysiology and Transplantation, University of Milan, Milan, Italy

$\overline{EP}$	Average external power
$\left[\overline{EP}\right]$	Average external power *per player*
EP_{ASP}	Overall external power corresponding to the individual ASP
EP_{thr}	External power threshold
ES	Cohen's d effect size
EW	External work
FB	Fullback
GNSS	Global Navigation Satellite Systems
GPS	Global Positioning System
highEW	External work performed over the EP_{thr}
$highEW_{force}$	External work performed over the EP_{thr} when velocity is between 0 and 1/3 v_0
$highEW_{power}$	External work performed over the EP_{thr} when velocity is between 1/3 v_0 and 2/3 v_0
$highEW_{speed}$	External work performed over the EP_{thr} when velocity is between 2/3 v_0 and v_0
MID	Midfielders
MP	Metabolic power
$\overline{MP}$	Average metabolic power
$\left[\overline{MP}\right]$	Average metabolic power *per player*
n	Number of athletes
P	Player
ST	Striker
t	Time
TD	Total distance
$\left[\overline{TD}\right]$	Average total distance *per player*
$\overline{v}$	Average velocity
$[\overline{v}]$	Average velocity *per player*
v_i	Starting velocity
v_t	Instantaneous velocity
v_0	X-intercept of the individual ASP
$\dot{V}O_2max$	Maximal oxygen consumption
$\dot{V}O_2$	Actual oxygen consumption
WG	Winger
WM	Wide midfielder

Introduction

Training load assessment is an essential prerequisite to manage at best players' health, physical fitness and wellness. The use of tracking technologies, both during matches and training sessions, has become common practice not only in professional sports. Indeed, information collected by means of these systems allows one to understand the amount of work done and the relative intensities through a multitude of performance indicators. The real challenge consists in identifying the parameters providing a clear picture of the performance, which is not an easy job (!), because some widespread indicators/metrics are inherently limited. The following paragraphs describe this state of affairs, as well as some possible solutions to overcome these limitations, regarding more specifically mechanical and metabolic power assessment. For the sake of simplicity, following a general introduction, match data from a professional soccer team will be presented and discussed. In particular, we will deal with three common indicators of the players' performances: (i) speed and acceleration alone, (ii) mechanical and (iii) metabolic power as described in some detail below. Even if these data will be centred on soccer, the general considerations deriving therefrom can also be extended to most other major team sports involving erratic fluctuations between high intensity running actions and lower intensity locomotion.

Locomotion workload

Since the late 1990s and early 2000s, tracking technologies to measure athletes' activity have been increasingly used as a source of information to analyse performances by measuring players' movements (Schutz and Chambaz 1997; Schutz and Herre 2000; Cummins et al. 2013). The instantaneous velocity assessed from the position (i.e. the derivative of the displacement with respect to time) by means of optical tracking or Global Navigation Satellite Systems (*GNSS*), or directly measured from the Doppler shift with *GNSS*, and hence the overall distance covered has been used from the very beginning as a global indicator of players' workload (Aughey 2013; Barnes et al. 2014; Bradley et al. 2013, 2016; Di Salvo et al. 2009, 2013; Grey and Jenkins 2010). However, the distance per se cannot provide any information on *how* it has been covered since, in spite of the strong correlation between distance and energy expenditure, an inevitable increase in energy cost can be observed the greater the amount of accelerated running (Osgnach et al. 2010). Indeed, the energy cost corresponding to the top acceleration of an elite soccer player at the very beginning of a maximal sprint can be more than 10 times larger than that of running at constant speed on flat terrain (di Prampero et al. 2005). For this reason, it would be preferable to use energy expenditure instead of distance as a measure of the total volume, thus avoiding to overlook even the smallest increase in the energy cost (e.g. Gaudino et al. 2013).

An alternative approach, but equally effective, is to refer to the external work instead of the energy expenditure: also in this case, there would be the possibility to consider the work done to move the mass of the player at a certain velocity, as well as the increase in kinetic energy due to the acceleration.

The foregoing description has focused on determining the overall workload volume that, once normalized by the time (t), yields the intensity of the exercise (see Table 1).

Table 1 Performance indicators commonly used in team sports

	Volume	Intensity
Locomotion	*TD* total distance [m] [km]	$\bar{v}$ average velocity [m s^{-1}] [km h^{-1}] [m min^{-1}]
Metabolic	*EE* energy expenditure [J kg^{-1}] [kcal kg^{-1}] [ml O_2 kg^{-1}]	$\overline{MP}$ average metabolic power [W kg^{-1}] [kcal kg^{-1} min^{-1}] [ml O_2 kg^{-1} min^{-1}]
Mechanical	*EW* external work [J kg^{-1}]	$\overline{EP}$ average external power [W kg^{-1}]

The central column (volume) refers to the overall performance (distance, *TD*; energy expenditure, *EE*; external mechanical work, *EW*); when referred to the corresponding time, the normalised indexes reported in the right column (intensity) can also be obtained (average velocity, $\bar{v}$; average metabolic power, $\overline{MP}$; average external power, $\overline{EP}$). Common units for expressing the indicated variables are also reported

Regardless of the performance indicators adopted to assess the overall volume and/or intensity, the particular nuances of the performance are difficult to understand without a deep analysis of the most demanding activities that the players carry out on the field. And that is why, from the first use of tracking systems, the analysis of the speed zones has been widely used by emphasising the association between high-speed distance covered and high-intensity efforts (Aughey 2013; Barnes et al. 2014; Bradley et al. 2013, 2016). It is indeed true that the higher the constant speed the higher the oxygen consumption and the external power, once a steady state is attained (Cavagna and Kaneko 1977; Cavagna et al. 1976). However, basing the analysis solely on the running velocity completely ignores the key role of acceleration and deceleration in affecting the exercise demands (di Prampero et al. 2005; accompanying paper, di Prampero et al. 2023). Therefore, the acceleration (positive and negative, i.e. the second derivative of the displacement with respect to time) has become an additional parameter investigated for the comprehension of the performance. In this case too, the analysis procedure is very similar to that used for the velocity: defining one or more thresholds, counting the actions carried out above a threshold and measuring the time spent and the distance covered in each acceleration zone (Harper et al. 2019; Vanrenterghem et al. 2017).

This manuscript is focused on the role of acceleration in increasing both metabolic and mechanical power; the use of acceleration alone, in fact, cannot provide a clear picture of the higher forces acting on the body of the player, especially when related to the individual acceleration capabilities, unless considering at one and the same time also the running velocity. The following paragraphs are devoted to describe in some detail the interplay between velocity and acceleration, discussing the advantages of taking into account both metabolic and mechanical power and showing how, in practice, they can be used to describe the players' performance in a real-world context (i.e. a full season of official matches).

Count the accelerations: what are we actually counting?

Since the beginning of tracking data analysis, as mentioned above, acceleration is one of the basic parameters used to describe the most challenging phases (Harper et al. 2019). Indeed, one of the most widespread performance indicators, in this specific case, is the number of intense accelerations (#*acc*). To obtain this information, the following two criteria should be met: the acceleration must (i) exceed a defined threshold (e.g. $a > 2.5$ m s^{-2}) and (ii) be maintained above that level for at least a minimum time (e.g. $t > 0.5$ s). Once these two conditions are met, one acceleration event is counted. In addition to the number of efforts (#*acc*), time spent and distance covered above the threshold can also be used as complementary information. Although easy to calculate, #*acc* can be really misleading: indeed, its aim is to describe the most intense neuromuscular efforts, a fact, however, that cannot be achieved dealing with the acceleration alone. To grasp the whole concept here, it's useful to refer to Sonderegger et al. (2016) who tested 21 young soccer players performing four maximal sprints over 50 m, starting from different velocities: standing (0 km h^{-1}), walking (6 km h^{-1}), jogging (10.8 km h^{-1}) and running (15 km h^{-1}). These data show that the time course of the velocity during the sprints follows an exponential time course regardless of the starting speed. More specifically, the increase of the velocity at time t (Δv_t), over and above the starting one (v_i) is described by:

$$\Delta v_t = (v_0 - v_i) \cdot (1 - e^{-(t/\tau)}) \quad (1)$$

where v_0 is the maximal velocity attained during the sprint and τ the time constant of the process. Hence, the instantaneous velocity (v_t) throughout the whole acceleration phase is described by:

$$v_t = v_i + \Delta v_t = v_i + (v_0 - v_i) \cdot (1 - e^{-(t/\tau)}) \quad (2)$$

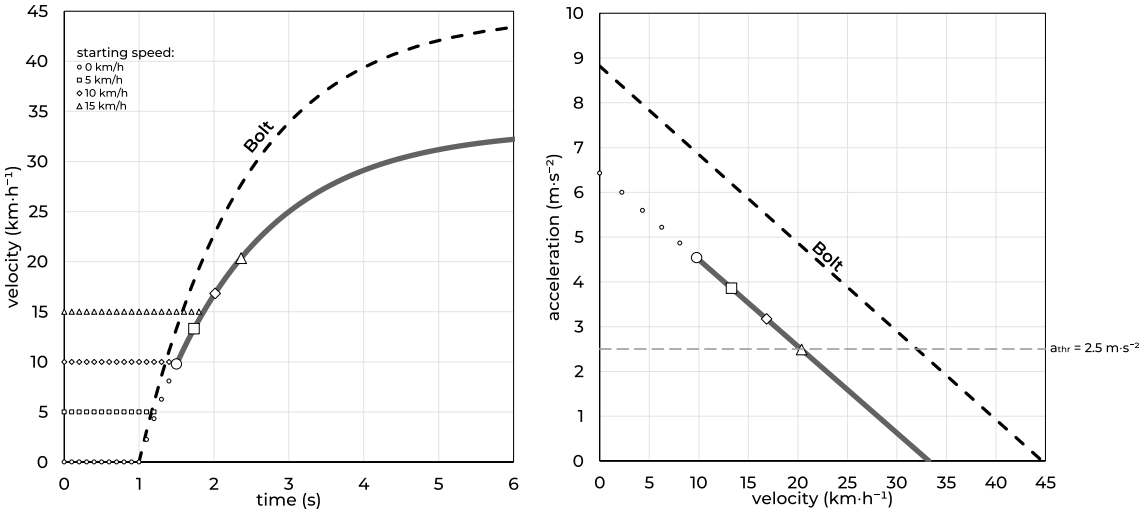

Fig. 1 Left panel: modelled time course of the velocity during four maximal sprints (assuming v_0 and τ equal to 33.23 km h^{-1} and 1.46 s, respectively) starting either from a standing position (grey circles) or from running at constant increasing velocities: 5 km h^{-1}, 10 km h^{-1} and 15 km h^{-1} by a high-level soccer player. The time course of the velocity, neglecting the initial phase, was essentially equal in all the conditions (black curve). Large empty symbols indicate the time at which the acceleration peak was achieved: starting from 0 km h^{-1} (open circle), from 5 km h^{-1} (open square), from 10 km h^{-1} (open diamond) and from 15 km h^{-1} (open triangle). Time course of the velocity from the extant 100-m world record of Usain Bolt (9.58 s, black dashed curve, from di Prampero et al. 2021) is also shown. Right panel: the instantaneous forward acceleration, as obtained from the first-time derivative of the curves reported in the left panel, is plotted as a function of the corresponding velocity. Large empty symbols indicate the peak acceleration attained from any given starting velocity: 0 km h^{-1} (open circle), 5 km h^{-1} (open square), 10 km h^{-1} (open diamond) and 15 km h^{-1} (open triangle). Thick dashed line refers to Usain Bolt 100-m world record. The thin grey dashed horizontal line, corresponding to an acceleration of 2.5 m s^{-2}, shows that, when running at velocities > 20 km h^{-1}, this acceleration threshold can neither be achieved nor surpassed by high-level soccer player

Obviously enough, when starting from a standing position, i.e. when $v_i = 0$, this equation reduces to the one reported in several papers describing this state of affairs (di Prampero et al. 2005, 2021; Samozino et al. 2016):

$$v_t = v_0 \cdot (1 - e^{-(t/\tau)}) \tag{3}$$

It also follows that, in all cases, the instantaneous acceleration (a_t) is described by:

$$a_t = (v_0 - v_t)/\tau \tag{4}$$

Equations 2 and 4 allowed us to approximate the maximal acceleration achievable by an elite soccer player starting from four different velocities (0, 5, 10 and 15 km h^{-1}), assuming v_0 and τ equal to 33.23 km h^{-1} (9.23 m s^{-1}) and 1.46 s, respectively (as reported below, see section Materials and Methods). A time delay after the beginning of the acceleration phase (≈ 0.5 s), necessary to reach the acceleration peak, was also considered to take into account (i) the time needed for the maximal muscle activation, (ii) the filtering of the raw velocity signal to avoid the intra-step acceleration/deceleration and (iii) the specific conditions related to soccer context, obviously substantially different from a 100-m sprint. By doing so, the Acceleration-Speed Profile (*ASP*) and the theoretical maximal accelerations calculated are within the range of those reported by Sonderegger et al. (2016). In fact, as shown in Fig. 1 (left panel), although the same top velocity (v_0 = 33.23 km h^{-1}) is attained in all cases, the acceleration peak decreases with increasing the speed wherefrom the sprint is initiated. It goes without saying that, as indicated in Fig. 1 (right panel), a single acceleration threshold (e.g. a_{thr} = 2.5 m s^{-2}) represents a different fraction of the best acceleration achievable over the four circumstances described above. For example, the a_{thr} is about 55% of the acceleration peak when starting from a standstill, up to almost 100% of the maximal acceleration achievable when the subject starts sprinting from 15 km h^{-1}.

It's precisely for this reason that the total number of #*acc* inevitably includes a multitude of events that could also be very different from each other in terms of intensity. For instance: one that, starting from low velocities, although overcoming the threshold, requires only a fraction of the maximal acceleration achievable as compared to another one that, starting from higher velocities, although attaining the maximal possible acceleration, might not be counted simply because it does not overcome a_{thr}. As a consequence, it is questionable whether this way of counting events effectively allows the coaching staff to obtain, at the end of a match, a clear picture of the intense actions performed by the players. Is there a solution for this?

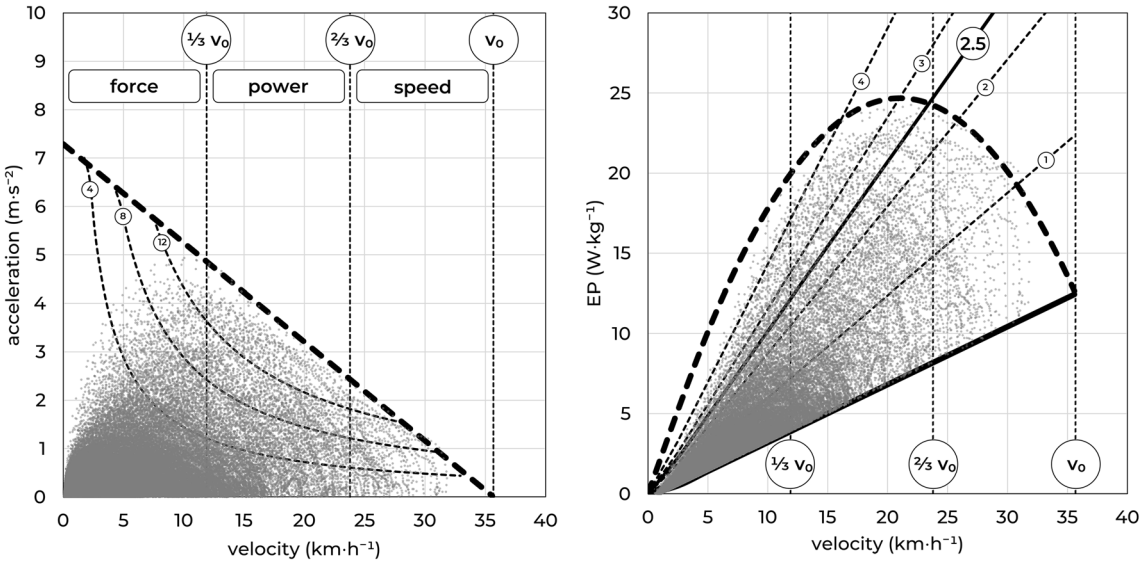

Fig. 2 Left panel: the Acceleration-Speed Profile (*ASP*), as obtained in-situ for a typical soccer player, is shown by the thick dashed line. The three hyperbolae (thin dashed lines) report the iso-mechanical power values of 4, 8 or 12 W kg^{-1} above that required for running at constant speed, respectively, necessary for accelerating as indicated. For example, when running at 1/3 v_0, an increase in mechanical power of 8 W kg^{-1} yields an acceleration of about 2.5 m s^{-2}, whereas the same increase in mechanical power when running at 2/3 v_0 yields a substantially lower acceleration value (i.e. about 1.2 m s^{-2}). Right panel: the mechanical power required for running at constant speed, including that to overcome the air resistance (W kg^{-1}), is shown by the thick continuous line, whereas the thin dashed lines report the overall mechanical power, including that for accelerating at 1, 2, 3, or 4 m s^{-2}, respectively. The thick dashed (dome-shaped) line is the maximal mechanical power that the subject can develop at the velocity indicated on the abscissa. The a_{thr} of 2.5 m s^{-2} is also shown. In both panels the *high-force, high-power* and *high-speed* zones are separated by the thin vertical dashed lines

Mechanical power approach

Given the interplay between velocity and acceleration (see accompanying paper, di Prampero et al. 2023), any acceleration carried out during a training session or a match, for each velocity value, can also be expressed using external power (*EP*). We would like to stress at this stage that, since at present it is difficult to calculate internal power in field conditions, in spite of its importance (Pavei et al. 2019; accompanying paper, di Prampero et al. 2023), the following analysis will be based on *EP* only.

In this framework, the first step is to define the maximal acceleration capacity throughout the player's sprint test (i.e. the maximal acceleration that can be achieved from any given initial speed), thus obtaining the individual force–velocity profile that may be suitable for this purpose (Cross et al. 2017). More recently, especially when dealing with team sports, an alternative method has been proposed based on the analysis of large amounts of *GPS* data gathered day by day over several football sessions: the Acceleration-Speed Profile in-situ (*ASP*) (Morin et al. 2021). Thanks to players' profiling via *ASP*, it is possible to know the highest forward acceleration achievable for each athlete over the entire running velocity spectrum (see continuous black line in Fig. 1), thus calculating the corresponding external power (EP_{ASP}). This allows one to abandon a fixed acceleration threshold, with the limits described above, and to refer to a fraction of EP_{ASP}. Margaria et al. (1971) showed that the maximal muscular power output decreases by about 20% when the subjects were invited to run at top speed up an ordinary flight of stairs, without interruption, from a $\dot{V}O_2$max steady-state exercise. Therefore, it would seem appropriate to use 80% of EP_{ASP} (EP_{thr}) as the individual threshold to identify the intense activities performed close to the greatest power capacity of the subject.

The second step of this process is to quantify the amount of external work carried out when *EP* is greater than EP_{thr}, i.e. to estimate the workload performed when high neuromuscular demands are required, regardless of the corresponding speed. In this regard, the overall external work per unit body mass is the time integral of *EP* (*EW*, J kg^{-1}), the fraction of which performed at high-intensity (i.e. $EP > EP_{thr}$) allowed us to define *highEW*. Running across the entire velocity spectrum, from 0 to v_0, occurs rather frequently for players involved in team sports and can be very different depending on specific exercises (e.g. small-sided games). Therefore, for a better comprehension of the performance, *highEW* can be divided into three components (Fig. 2, left panel), by splitting the whole velocity range into equal zones characterised by a prevalence of force, power or speed (Cahill et al. 2019):

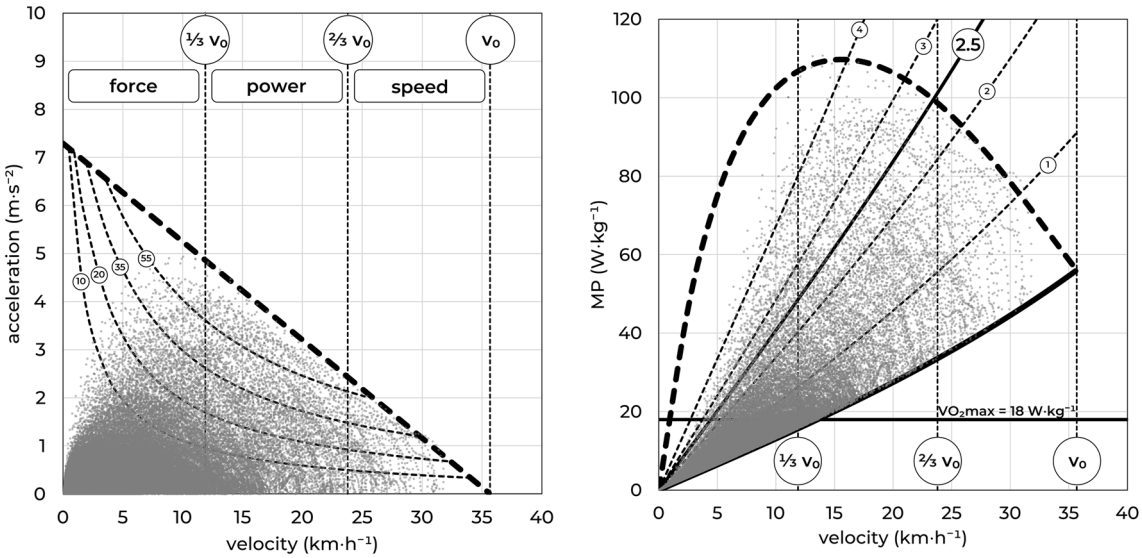

Fig. 3 Left panel: the Acceleration-Speed Profile (*ASP*), as obtained in-situ for a typical soccer player, is shown by the thick dashed line. The four hyperbolae (thin dashed lines) report the iso-metabolic power values of 10, 20, 35 or 55 W kg^{-1} above that required for running at constant speed, respectively, necessary for accelerating as indicated. For example, when running at 1/3 v_0, an increase in metabolic power of 55 W kg^{-1} yields an acceleration of about 3.6 m s^{-2}, whereas the same increase in metabolic power when running at 2/3 v_0 yields a substantially lower acceleration value (i.e. about 2.1 m s^{-2}). Right panel: the metabolic power required for running at constant speed, including that to overcome the air resistance (W kg^{-1}), is shown by the thick continuous line, whereas the thin dashed lines report the overall metabolic power including that for accelerating at 1, 2, 3, or 4 m s^{-2}. The thick dashed (dome-shaped) line is the maximal metabolic power that the subject can develop at the velocity indicated on the abscissa. The a_{thr} of 2.5 m s^{-2}, together with the metabolic power corresponding to a maximal oxygen uptake value of 18 W kg^{-1} (≈ 52 ml O_2 kg^{-1} min^{-1} above resting) are also shown. In both panels the *high-force, high-power* and *high-speed* zones are separated by the thin vertical dashed lines

a. $highEW_{force}$, when velocity is between 0 and 1/3 v_0;
b. $highEW_{power}$, when velocity is between 1/3 v_0 and 2/3 v_0;
c. $highEW_{speed}$, when velocity is between 2/3 v_0 and v_0.

Finally, we would like to stress that the number of events carried out above EP_{thr} is an extremely useful information for the coach. Indeed, it provides the possibility to count all the intense activities that lead the player in close proximity to his absolute limit thus avoiding the use of a fixed acceleration threshold (a_{thr}) which corresponds to very different external power values (see Fig. 2, right panel). Intense mechanical actions (#*burst*) are here defined as the number of bouts performed at external power values exceeding EP_{thr}. As such, the total number of #*burst* allows one to obtain a series of details providing a description thereof: duration (s), distance (m), starting speed (m s^{-1} or km h^{-1}), overall external work per unit body mass (J kg^{-1}), peak and average external power (W kg^{-1}).

Metabolic power approach

Something similar happens also when dealing with the energetic approach (di Prampero and Osgnach 2018; Polglaze and Hoppe 2019): any acceleration performed from a given velocity value requires an additional metabolic power output (*MP*, W kg^{-1}), above that for running at constant speed, indicated by the iso-power curves in Fig. 3 (left panel). In this case, the sudden increase of *MP* requirement over and above the player's actual $\dot{V}O_2$ is fulfilled by the immediate intervention of anaerobic stores (mainly phosphocreatine breakdown), the most demanding phases being represented by spells in which the *MP* exceeds the subject's $\dot{V}O_2$max. Hence, the energetic approach makes it possible to assess the overall energy expenditure (*EE*, J kg^{-1}), the fraction derived from anaerobic sources (*AnE*, J kg^{-1}), the alternation between high demanding actions (Metabolic Power Events, #*MPE*) and

Table 2 Performance indicators referring to three categories reported in Table 1

	Locomotion	Mechanical	Metabolic
1	*TD* total distance [m] [km]	*EW* external work [J kg^{-1}]	*EE* energy expenditure [J kg^{-1}] [kcal kg^{-1}] [ml O_2 kg^{-1}]
2	$\bar{v}$ average velocity [m s^{-1}] [km h^{-1}] [m min^{-1}]	$\overline{EP}$ average ext power [W kg^{-1}]	$\overline{MP}$ average met power [W kg^{-1}] [kcal kg^{-1} min^{-1}] [ml O_2 kg^{-1} min^{-1}]
3	$D_{speed>5.5}$ distance > 5.5 m s^{-1} [m]	*highEW* high ext work [J kg^{-1}]	*AnE* anaerobic energy [J kg^{-1}] [kcal kg^{-1}] [ml O_2 kg^{-1}]
4	$D_{speed>7.0}$ distance > 7.0 m s^{-1} [m]	$highEW_{force}$ high ext work 0–1/3v_0 [J kg^{-1}]	$D_{MP>20}$ distance > 20 W kg^{-1} [m]
5	#*speed* speed events > 7.0 m s^{-1} [#]	$highEW_{power}$ high ext work 1/3v_0–2/3v_0 [J kg^{-1}]	AnE/EE anaerobic energy fraction [%]
6	$D_{acc>2.5}$ distance > 2.5 m s^{-2} [m]	$highEW_{speed}$ high ext work > 2/3v_0 [J kg^{-1}]	$D_{MP>20}/TD$ distance > 20 W kg^{-1} fraction [%]
7	#*acc* acceleration events > 2.5 m s^{-2} [#]	#*burst* bursts events $EP > EP_{thr}$ [#]	#*MPE* metabolic power events $MP > \dot{V}O_2$ [#]

Left column (locomotion): lines one and two, as from Table 1; lines three to five: distance covered at velocity greater than 5.5 or 7.0 m s^{-1} and number of occasions in which the velocity exceeded 7.0 m s^{-1}; lines six and seven: distance covered at accelerations greater than 2.5 m s^{-2} and number thereof. Central column (mechanical): lines one and two, as from Table 1; line three: external mechanical work performed when *EP* is greater than EP_{thr}; lines four to six: external mechanical work performed above EP_{thr} while running within the indicated speed intervals; line seven: number of occasions in which *EP* exceeded EP_{thr} (i.e. 80% of the mechanical power corresponding to the thick dome-shaped dashed line of Fig. 2). Right column (metabolic): lines one and two, as from Table 1; line three: time integral of the instantaneous metabolic power above the corresponding estimated $\dot{V}O_2$; line four: overall distance covered at a metabolic power greater than 20 W kg^{-1}; line five: ratio between net anaerobic energy contribution and overall energy expenditure; line six: ratio between the distance covered at a metabolic power greater than 20 W kg^{-1} and the total distance; line seven: number of occasions in which the metabolic power exceeded the corresponding estimated $\dot{V}O_2$

recovery phases and their characteristics in terms of duration and average *MP* (Osgnach and di Prampero 2018) (see Table 2). This description of the performance is profoundly different from the previous mechanical one and, as such, it should be considered as a complementary view. Indeed, especially referring to the high-intensity bouts, dealing with the mechanical approach means to describe the events held through high neuromuscular efforts (#*burst*), whereas the metabolic approach deals with the most demanding actions (#*MPE*) which are carried out thanks to the utilisation of the anaerobic energy stores and are followed by recovery periods in which a fraction of the O_2 debt is paid at the expense of the oxidative processes (Osgnach and di Prampero 2018). Thus, whereas all #*burst* are #*MPE* in so far as they require large amounts of metabolic power, the reverse is not true, since #*MPE* may be spread over relatively long time periods and, as such, they do not necessarily belong to the #*burst* "family".

The three different indicators described above allowed us to obtain data on the players (i) who completed the entire match as compared to the substitutes, (ii) their role, as well as (iii) the time course of the maximal mechanical and metabolic power events and (iv) the average duration of the corresponding performance and recovery times over subsequent 15-min periods throughout the overall match.

The relevant data are reported and discussed in the subsequent sections of the manuscript.

Materials and methods

38 official match performance data were collected from a professional soccer team competing in the Italian second league (Serie B) during the season 2020–2021. A total of 549 observations were considered from 29 individual field players across the entire season (mean ± SD; age = 26.6 ± 4.4

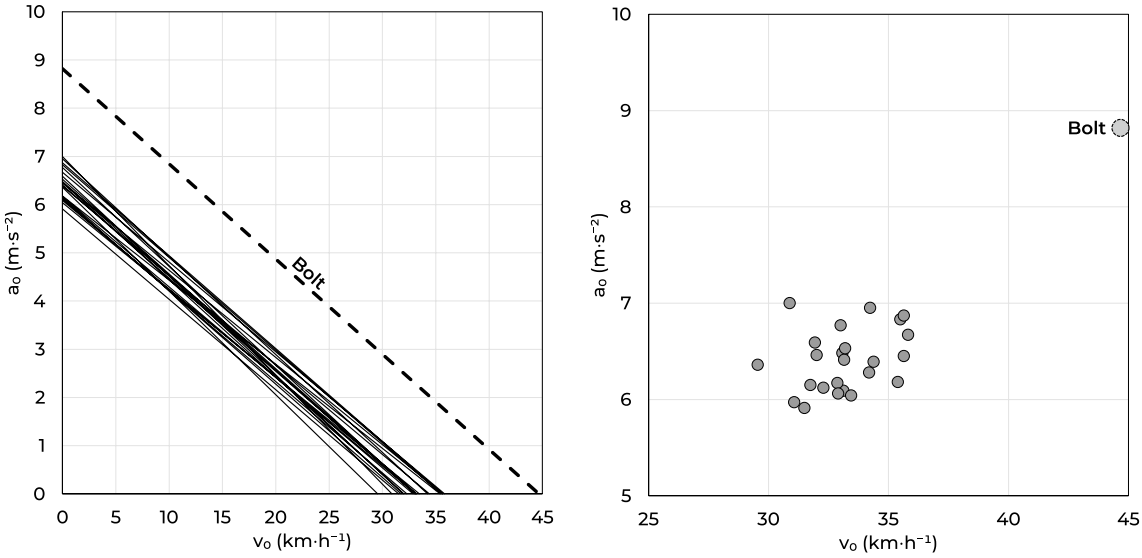

Fig. 4 Left panel: individual Acceleration-Speed Profiles (*ASP*) of the 24 soccer players (continuous black lines) and of Usain Bolt (dashed line, from di Prampero et al. 2021). Right panel: corresponding v_0 and a_0 values (dark full dots, soccer players; grey circle, Usain Bolt)

yr; mass = 74.5 ± 5.0 kg; stature = 1.82 ± 0.06 m) who played in different positions, regardless whether or not they completed the overall match: Centre Backs, *CB* (n = 5; observations = 88); Fullbacks, *FB* (n = 5; observations = 89); Centre Midfielders, *CM* (n = 5; observations = 66); Wide Midfielders, *WM* (n = 5; observations = 139); Wingers, *WG* (n = 6; observations = 115); Strikers, *ST* (n = 3; observations = 52). Goalkeepers were excluded from the study. Players' movements were collected using a *GPS* system (gpexe Pro^2, firmware v. 58, Exelio srl, Udine, Italy) at a sampling rate of 18.18 Hz and processed using the proprietary software (gpexe web app v. 7.4.66). The device has been previously determined as reliable and valid for the distances and the sprint mechanical properties in question as reported by Hoppe et al. (2018).

As mentioned above, we analysed three common indicators of the players' performance: (i) locomotion (speed and acceleration), (ii) external mechanical power and (iii) metabolic power. For each category, only the performance indicators reported in Table 2 were analysed, as briefly described below.

More specifically, as concerns the category "locomotion", we determined the overall distance covered and the corresponding average velocity, as well as the distance covered at speeds above two specific thresholds (5.5 and 7.0 m s^{-1}) together with the number of events in which the speed exceeded the higher threshold. In addition, we assessed also the distance covered at an acceleration ≥ 2.5 m s^{-2} and the number of events thereof.

As concerns the category "mechanical", a basic prerequisite has been the analysis of the individual in situ *ASP*s for each match, as obtained on the basis of the following criteria: (i) duration ≥ 45 min, and (ii) maximum speed attained ≥ 80% of v_0 (Morin et al. 2021; Clavel et al. 2023). A total of 320 in situ *ASP*s met the requirements and were obtained from 24 different players. The individual profile utilised to calculate the most intense neuromuscular activities was estimated from the average v_0 and a_0 of all the available profiles for each subject. For the 5 players on whom no data coherent with the above criteria were available, the corresponding *ASP* was obtained from the average v_0 and a_0 for players of the same role. The average v_0 and a_0 were 33.23 ± 1.55 km h^{-1} and 6.43 ± 0.30 m s^{-2}, respectively (Fig. 4).

This allowed us to determine the overall external mechanical work and the corresponding average power, as well as the external mechanical work performed above a predetermined threshold (EP_{thr}) equal to 80% of the maximal mechanical power corresponding to the thick dome-shaped dashed line of Fig. 2, together with the external mechanical work performed in three selected speed zones (i.e. between 0 and 1/3 v_0; between 1/3 v_0 and 2/3 v_0; between 2/3 v_0 and v_0). In addition, the number of events in which the external mechanical power exceeded EP_{thr} was also determined.

As concerns the category "metabolic", we determined the overall energy expenditure and the corresponding average metabolic power, as well as the amount of energy derived from anaerobic source (high energy phosphates breakdown and/or lactate accumulation), as obtained from the time integral of the instantaneous metabolic power above the corresponding time integral of the estimated $\dot{V}O_2$. The overall amount of energy spent at a metabolic power greater that the subjects' $\dot{V}O_2$max above resting

Table 3 Team match data (means ± SD, and corresponding minimum and maximum values) of the performance indicators described in Table 2, as obtained from 38 official matches

Locomotion	Mechanical	Metabolic
TD 105,965 ± 3198 m 98,950 – 114,215	*EW* 112,360 ± 4466 J kg^{-1} 100,129 – 124,209	*EE* 482,948 ± 17,343 J kg^{-1} 436,149 – 526,497
$[\overline{v}]$ 6.68 ± 0.18 km h^{-1} 6.33 –7.07	$[\overline{EP}]$ 1.97 ± 0.07 W kg^{-1} 1.78 – 2.13	$[\overline{MP}]$ 8.46 ± 0.28 W kg^{-1} 7.77 – 9.05
$D_{speed>5.5}$ 7654 ± 871 m 5914 – 9848	*highEW* 5193 ± 513 J kg^{-1} 4046 – 5973	*AnE* 191,035 ± 6743 J kg^{-1} 172,593 – 203,214
$D_{speed>7.0}$ 1566 ± 321 m 950 – 2192	$highEW_{force}$ 1085 ± 138 J kg^{-1} 807 – 1400	$D_{MP>20}$ 27,343 ± 1660 m 22,933 – 30,370
#*speed* 100.7 ± 17.7 68 – 143	$highEW_{power}$ 2756 ± 312 J kg^{-1} 2095 – 3369	*AnE/EE* 0.401 ± 0.016 0.364 – 0.447
$D_{acc>2.5}$ 1028 ± 104 m 820 – 1246	$highEW_{speed}$ 1359 ± 212 J kg^{-1} 922 – 1812	$D_{MP>20}/TD$ 0.247 ± 0.031 0.151 – 0.327
#*acc* 284.6 ± 35.5 220 – 373	#*burst* 504.4 ± 46.0 394 – 593	#*MPE* 1588 ± 82 1360 – 1735

As described in the "Materials and methods" section, the *per player* match average values were obtained dividing by 10 the appropriate team match values

(assumed to be 18 W kg^{-1})[1] was also calculated, together with the number of occasions in which this actually occurred. In addition, we will also calculate two indices widely used to obtain information on the overall anaerobic energy yield: i.e. (i) the ratio between net anaerobic energy contribution and overall energy expenditure (AnE/EE) and (ii) the ratio between the distance covered at a metabolic power greater than 20 W kg^{-1} and the total distance ($D_{MP>20}/TD$).

A summary of the so obtained performance indicators is reported in Table 2.

Data collection for this retrospective analysis was performed within the typical official match context, under usual technical and medical supervision. No specific intervention was therefore required for this study, which was performed according to Declaration of Helsinki.

All results are presented as means ± SD unless otherwise stated. Where appropriate, in addition to team and individual data, *per player* match average was calculated as the sum of the activity carried out by all the athletes (n) who took part in each single match (i.e. starters + substitutes) divided by 10. As an example, the match's average total distance *per player* $\left(\left[\overline{TD}\right]\right)$ is defined as follows:

$$\left[\overline{TD}\right] = \frac{\sum_{P=0}^{P=n} TD_n}{10} \tag{5}$$

Similarly, the intensity performance indicators related to average *per player (P)* match were calculated accordingly. Thus, the average velocity *per player* ($[\overline{v}]$) is calculated as:

$$[\overline{v}] = \frac{\left[\overline{TD}\right]}{\sum_{P=0}^{P=n} t_n} \tag{6}$$

Indeed, rather than considering the athletes who completed the entire match (who are often those with the lowest workload performance), this seems to be a more appropriate criterion for expressing the match player's average, especially after the introduction of five substitutions. One-way or two-way ANOVA with Bonferroni post hoc test was used to identify the statistically significant differences on mechanical and metabolic variables vs. position, starting speed and time. The Pearson statistical test was used to analyse the correlations between the examined variables. A p-value < 0.05 was considered statistically significant. All analyses were performed using GraphPad Prism. Finally, if required, the Cohen's d effect size was calculated to determine the difference between match periods and presented with 95% lower and upper confidence intervals ($ES \pm 95\%\ CI$): magnitudes were defined as trivial (≤ 0.2), small (> 0.2–0.6), moderate (> 0.6–1.2) and large (> 1.2) (Batterham and Hopkins 2006).

[1] The oxygen consumption rate has been converted to W kg^{-1} (= J s^{-1}·kg^{-1}) assuming an energetic equivalent of 20.9 J ml O_2^{-1}, which is strictly true for a metabolic Respiratory Quotient = 0.955.

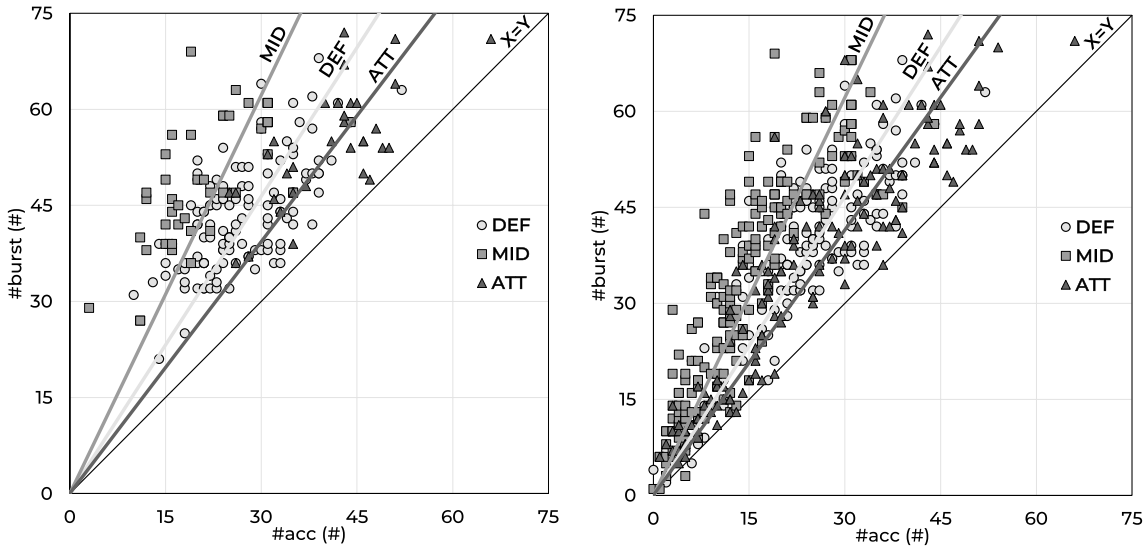

Fig. 5 Number of events in which the mechanical power exceeded the threshold, set at 80% of the external power corresponding to the subject's *ASP* (*#burst*), is plotted as a function of the number of events in which the acceleration exceeded 2.5 m s^{-2} (*#acc*). Regressions referring to DEF (open circle, light grey), MID (open square, dark grey) and ATT (open triangle, black), together with identity line, are also show. Left panel: players who completed the entire match only. Right panel: all players, substitutes included. See text for details

Results

The mean official match time was 95 min 23 s ± 1 min 58 s. Team match data performance indicators for locomotor, mechanical and metabolic approaches are presented in Table 3.

Locomotion

The average distance covered by all the players who completed the entire match (thus neglecting the intervening substitutes) was 10,277 ± 857 m, which is 3% lower than *per player* match reference value, as calculated from Table 3 (i.e.: 105,965/10 = 10,596.5 m). In fact, among the 214 athletes who played the whole game, 132 were defenders (DEF; CB + FB: 9862 ± 718 m), 49 midfielders (MID; CM + WM: 11,207 ± 553 m) and 33 attackers (ATT; ST + WG: 10,555 ± 457 m). Similarly, when attention shifts to high velocities and acceleration values (again neglecting the intervening substitutes), the average distances covered above 5.5 m s^{-1}, 7.0 m s^{-1} and 2.5 m s^{-2} by players completing the overall match were 703.8 ± 255.0 m, 148.7 ± 92.7 m and 98.1 ± 31.7 m, respectively. The distance covered in the upper speed and acceleration zones was achieved through 9.5 ± 5.2 speed and 27.2 ± 10.0 acceleration events. As mentioned above, the so obtained performance indicators are slightly lower than those obtained from the data reported in Table 3 that takes into account also the intervening substitutes.

Mechanical

As a first step of the analysis, individual in situ *ASP*s were obtained for each match (see Materials and Methods). The average external work calculated for players who completed the entire match (thus neglecting the intervening substitutes) was 10,780 ± 1292 J kg^{-1}, the fraction of which performed for *EP* greater than EP_{thr} was about 4.4%, equivalent to 475.6 ± 157.9 J kg^{-1}. This high intensity work has been carried out for approximately 50% in the *power* zone (250.5 ± 90.6 J kg^{-1}, i.e. velocity between 1/3 v_0 and 2/3 v_0), and for the remaining 50% equally divided into the *force* (104.5 ± 34.3 J kg^{-1}, i.e. velocity lower than 1/3 v_0) and *speed* (120.6 ± 66.4 J kg^{-1}, i.e. velocity higher than 2/3 v_0) zones. The average number of *#burst* performed during a match was 47.0 ± 11.1.

An analysis of Fig. 5 (left panel) highlights the difference between the events obtained from the "locomotion" (*#acc*) and the "mechanical" (*#burst*) approach. Even if, as a first approximation, one would expect to observe similar numbers, when referring to the players who complete the entire match, the number of *#burst* is about 60% greater than the number of *#acc* (slope = 1.61, R^2 = 0.930; P-value < 0.0001). It is interesting to note that this difference is smaller for ATT (slope = 1.36, R^2 = 0.977; P-value < 0.0001), higher for DEF (slope = 1.56, R^2 = 0.961; P-value < 0.0001) and very much higher for MID (slope = 2.30, R^2 = 0.946; P-value < 0.0001).

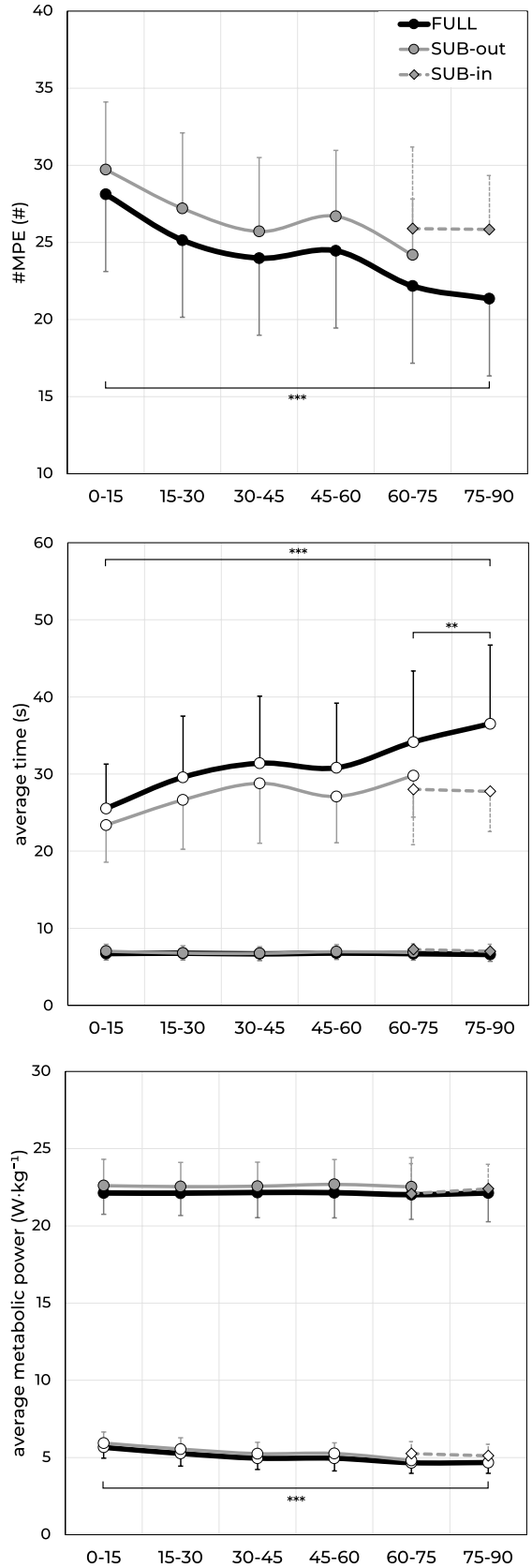

Fig. 6 Average number of *#MPE* (Upper panel), average work and recovery time (Middle panel) and average metabolic power of the work and recovery phases (Lower panel) in the indicated 15-min periods. Continuous lines refer to players who completed the entire match (FULL filled circle, black line), who have been replaced (SUB-out filled circle, grey line) or to substitutes involved in the two last periods (SUB-in filled diamond, dashed grey line). Full and empty symbols refer to the average values of work and recovery phases, respectively. SD is indicated by the vertical error bars. (***: $p < 0.0001$, **: $p < 0.001$, only for players who complete the entire match)

The data further show that these differences remain substantially unchanged if, in addition to the starters, the substitutes are also taken into account (see Fig. 5, right panel): all players (slope = 1.64, $R^2 = 0.924$; P-value < 0.0001), only ATT (slope = 1.45, $R^2 = 0.957$; P-value < 0.0001), only DEF (slope = 1.58, $R^2 = 0.958$; P-value < 0.0001) and only MID (slope = 2.22, $R^2 = 0.936$; P-value < 0.0001).

Metabolic

The average energy expenditure for all players estimated over the entire match (thus neglecting the intervening substitutes) was 46,416 ± 5000 J kg^{-1}. The differences concerning field position, although similar to those observed by using the locomotor approach, emphasize the specific behaviour of each role: defenders (DEF; CB + FB: 43,959 ± 4313 J kg^{-1}), midfielders (MID; CM + WM: 51,495 ± 3101 J kg^{-1}) and attackers (ATT; ST + WG: 48,705 ± 2444 J kg^{-1}). In terms of high-demanding activities from a metabolic perspective, AnE/EE, for athletes completing the entire match (often those with the lowest workload values, Osgnach and di Prampero 2018), was 0.401 ± 0.016 and ranges from a minimum of 0.364 to a maximum of 0.447. On the contrary, the ratio $D_{MP>20}/TD$, a widespread index the aim of which is to yield information on the overall anaerobic energy supply, was significantly lower and equal to 0.247 ± 0.031, ranging from a minimum of 0.151 to a maximum of 0.327. Indeed, the relationship between the above two indices was weak ($D_{MP>20}/TD = -0.42 \cdot AnE/EE + 0.42$; $R^2 = 0.049$; P-value = 0.001) and the negative slope even showed that $D_{MP>20}/TD$ decreased when AnE/EE increased. Finally, the average number of *#MPE* performed during 90 min was 151.9 ± 19.8, which means that a high-intensity episode (average duration of 6.7 ± 0.5 s, minimal and maximal time 5.5 and 7.8 s, respectively; average metabolic power of 22.1 ± 1.2 W kg^{-1}, minimal and maximal metabolic power 20.1 and 25.6 W kg^{-1}, respectively) was followed by a recovery period of 31.4 ± 5.5 s (minimal and maximal time ranges were 21.1 and 52.7 s, respectively) carried out at an average metabolic power of 4.96 ± 0.52 W kg^{-1} (minimal and maximal average metabolic power ranges from 3.56 and 6.29 W kg^{-1}, respectively). Thus, on the average, an overall cycle (work + recovery) occurs every ≈ 37 s.

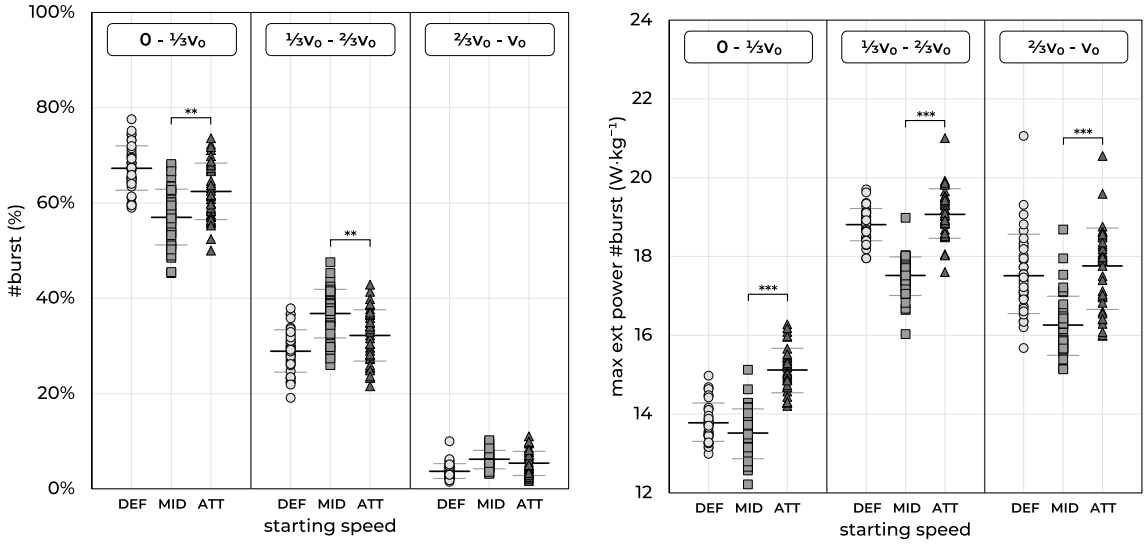

Fig. 7 Percentage (Left panel) and corresponding peak external power (Right panel) of #*burst* performed by DEF (open circle), MID (open square) and ATT (open triangle) starting from the low (0–1/3 v_0), intermediate (1/3 v_0–2/3 v_0) and high (2/3 v_0–v_0) speed zones. Data report the grand average as obtained from all players involved throughout the entire 38 matches. For each category, averages ± SD are indicated by the thick and thin horizontal lines, respectively. (***: p < 0.0001, **: p < 0.001)

The evaluation of successive 15-min periods referred to the players who complete the entire match (Fig. 6, upper panel) shows that the number of #*MPE* decreases from 28.12 ± 4.52 in the first to 21.35 ± 4.69 in the last period (P-value < 0.0001; ES − 1.47; CI ± 0.21), indicating that the average recovery time between subsequent #*MPE* increases from 25.53 ± 5.76 s in the first to 36.52 ± 10.21 s in the last 15-min period (P-value < 0.0001; ES 1.32; CI ± 0.21) (see Fig. 6, middle panel). Furthermore, the intensity maintained during the recovery phases also tends to decrease from 5.66 ± 0.71 W kg^{-1} in the first to 4.67 ± 0.70 W kg^{-1} in the last 15-min period (P-value < 0.0001; ES − 0.69; CI ± 0.20) (Fig. 6, lower panel). Needless to say, both the increased recovery time and the decreased recovery intensity observed might be due not only to the onset of fatigue but also to the breaks needed to make the substitutions and/or the tactical conduct of the match that obviously affect the average recovery time reported above. Fatigue, however, seems to play a role in the effectiveness of the recovery; in fact, average recovery time from the last two 15-min periods (i.e. 60–75 and 75–90, see Fig. 6, middle panel) show that the substitutes who completed both these periods have no increase in average recovery time (28.02 ± 7.19 s vs. 27.75 ± 5.21 s) (P-value = 0.14; ES -0.04; CI ± 0.53) at variance with their teammates who played from the beginning of the match and are obviously subject to the same contextual conditions (34.14 ± 9.28 s vs. 36.52 ± 10.21 s) (P-value < 0.001; ES 0.25; CI ± 0.19).

Discussion

The preceding sections of this paper were devoted to an analysis of the interactions between velocity and acceleration with the aim of assessing players' performances in more detail. Actually, an isolated analysis of velocity or acceleration has inherent limitations, mainly when referring to acceleration per se. Indeed, mechanical and metabolic approaches, when related to the overall workload and average intensity, may appear somewhat redundant. However, when used to describe the most demanding activities, they provide valuable information on the nature of the exercise from the neuro-muscular mechanical perspective, on the one side, and the aerobic/anaerobic energy yield, on the other.

Mechanical

The number of high intensity mechanical events (#*burst*) was higher than the corresponding number of acceleration (#*acc*) when all the players who complete the entire match were considered (+ 61%). This dissimilar behaviour depends on the fact that the selected acceleration threshold (a_{thr} = 2.5 m s^{-2}) can be overcome only when the starting velocity is relatively low, thus reducing the possibility of attaining external mechanical power values greater than 80% of maximal *EP* (see Fig. 2, right panel). As such, it's crucially dependent on the different players' position as shown by the fact that, when considering the

players who complete the entire match, the ratio *#burst* / *#acc* for the ATT was 1.36, whereas for MID it was much larger, amounting to 2.30 (see Fig. 5). This state of affairs can originate from two main reasons. It is due to (i) the starting speed preceding a sprint, which is lower for ATT, that makes it easier to overcome a_{thr}, and higher for MID; indeed, a deeper analysis of *#burst* carried out in each position shows that ATT more often perform intense actions starting from low speed (i.e. velocity lower than 1/3 v_0) than MID (P-value < 0.001; ES 0.91; CI ± 0.47); on the opposite, MID more often perform intense actions starting from moderate speed (i.e. velocity between 1/3 v_0 and 2/3 v_0) than ATT (P-value < 0.001; ES 0.88; CI ± 0.47) (see Fig. 7, left panel). In addition, regardless of the starting speed in performing *#burst*, (ii) ATT express higher maximal external power compared to MID at low (P-value < 0.0001; ES 2.67; CI ± 0.62), moderate (P-value < 0.0001; ES 2.82; CI ± 0.63) and high velocities (P-value < 0.0001; ES 1.61; CI ± 0.52), a fact that per se makes it more likely to meet the conditions for counting also *#acc*, i.e. to overcome a defined threshold and to maintain the acceleration above that level for at least a minimum time (see Fig. 7, right panel).

Metabolic

The aim of the mechanical approach is to describe the most intense actions in terms of external power output, regardless whether they depend on a prevalence of force, power or speed. At variance, the metabolic analysis is to be preferred for a clearer interpretation of the sequence between high-demanding spells (i.e. utilisation of anaerobic energy stores) and the following recovery phases (i.e. O_2 debt paid at the expense of the oxidative processes), regardless of the status of the game (i.e. ball in play or not). These 'context-related' pauses are in any case useful to restore, fully or partially, the anaerobic stores showing that the description of match workloads based only on actual playing times is not appropriate. Indeed, the estimated recovery periods would be completely different once considering only the actual playing time, thus yielding an equivocal description of the sequence between high- and low-energy demanding phases (Linke et al. 2018; Rey et al. 2020). Interestingly, when focusing on high-intensity bouts, both the average duration and intensity of these phases remain almost unchanged from the beginning to the end of the match, for the players who completed the entire match as well as for the substitutes. In other words, this shows that the players are capable of carrying out highly demanding actions that last about 6–7 s (see Fig. 6, middle panel) at an average metabolic power of about 22 W kg^{-1} (see Fig. 6, lower panel) either at the beginning or the end of the match. Obviously enough, these findings can be attributed to the fact that the recovery time between high intensity phases increases progressively, thus leading to a consensual decline of about 20% of the overall energy expenditure from the beginning to the end of the match (i.e. from 8107 ± 1041 J kg^{-1} to 6532 ± 952 J kg^{-1} in the first and last 15-min period of the match, respectively, for the players who completed the entire match).

Conclusions

The general discussion reported above refers to entire official matches, as such it does not necessarily applies to different contexts (i.e. training sessions or specific exercises). However, it shows unambiguously that, as suggested in the accompanying paper devoted to the 100-m dash (di Prampero et al. 2023), in exercises characterised by frequent accelerated running phases, neither the velocity nor the acceleration alone can be considered appropriate indices of the corresponding mechanical or metabolic power. This state of affairs becomes of paramount importance in team sports, such as soccer, in which case traditional analyses based on velocity or acceleration alone yield partial information, strictly dependent on the specific players' roles and match characteristics. However, an analysis based on mechanical and metabolic power, combining at one and the same time velocity and acceleration, allows one to assess the players' performances with a substantially greater accuracy.

Furthermore, it seems crucial to highlight a substantial difference between the mechanical and metabolic approaches. Indeed, the latter yields information including both high-demanding phases and the subsequent recovery necessary to refill the energy stores. At variance, the mechanical approach is devoted to an analysis of the phases in which muscle activity increases "explosively" towards its maximum. Hence a combined and integrated analysis of the information provided by both approaches allows one to collect a more comprehensive view of the players' characteristics, thus programming at best matches and training strategies.

Author contribution statement PEdP, CO and GP conceived and designed research. CO, GP, JBM and PZ contributed analytical tools. PEdP, CO and GP analysed data. PEdP and CO wrote the manuscript. All authors read and approved the manuscript.

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Declarations

Conflict of interest CO and PEdP are scientific consultants for Exelio SRL, the company which provided the GPS units used in this study. Other authors have no conflict of interest to declare.

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